



Fazeel Nizam

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<https://fazeelnizam.github.io/portfolio/> FazeelNizam
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Profile

Innovative and self-driven Computer Engineer with hands-on experience in Embedded Systems, Edge AI, FPGA, and IoT. Proven ability to architect end-to-end systems, from firmware and hardware integration to cloud connectivity and computer vision pipelines. Seeking opportunities to design cutting-edge Embedded, IoT, AI/ML, and FPGA systems that bridge research and real-world applications.

Technical Skills

Embedded & Hardware, NVIDIA Jetson, Raspberry Pi, FPGA (Xilinx Artix-7/Spartan6), ESP32, Arduino

AI/ML, Computer Vision, OCR, Edge Computing, CNN, YOLO

Protocols, MQTT, LoRa, SDI-12, SPI, I²C, UART, BLE, HTTP/REST, JSON, RTSP

Programming & Scripting, Python, VHDL, Java, Embedded C/C++, JavaScript (ES6)

Software & Tools,

Vivado, MATLAB, Git, Proteus, AutoCAD, Figma, FreeRTOS, AWS IoT Core, CI/CD, Docker

Web Technologies,

ReactJS, NextJS, Node-RED, HTML5, CSS3, Tailwind, MUI, FastAPI, PostgreSQL, Nginx

Professional Experience

Trainee IoT Engineer, SLTMobitel Digital Projects Oct 2025 – Present | Colombo, Sri Lanka

- Developing an AI-powered parking management system by implementing low-latency RTSP video pipelines using GStreamer on WSL2 to bridge legacy CCTV infrastructure with modern AI processing.
- Benchmarked YOLOv11n and OCR inference performance on Raspberry Pi 5 and am currently evaluating the architectural feasibility of deploying to NVIDIA Jetson Orin Nano for production environments.
- Designing a scalable IoT ecosystem featuring smart access control, electrical power monitoring, and automated HVAC control driven by real-time CO₂, temperature, and humidity data analysis.

Technical Assistant, Career Guidance Unit, OUSL Mar 2025 – Sep 2025 | Nawala, Sri Lanka

- Engineered a multi-node environmental monitoring system and a portable Greenhouse Gas (GHG) Analyzer for agricultural research.
- Provide technical support for university-wide hybrid career events.

Trainee Electronic Engineer,

Arthur C Clarke Institute for Modern Technologies

Jul 2024 – Dec 2024

Katubedda, Sri Lanka

- Designed an ESP32-based Nano-Sterlite Orbital Magnetic Field Simulation Device (Helmholtz Cage), utilizing a PID algorithm to stabilize a frictionless air-bearing platform.

Technical Assistant, Neo Space Lab, OUSL

Sep 2023 – Dec 2023 | Nawala, Sri Lanka

- Designed UI/UX for a Research Publication Web App and IoT dashboards using NextJS and MUI.
- GitHub Repositories: *IoT Weight Measurement System* [🔗](#), *LKO Web Dashboard* [🔗](#)

Banking Trainee,

Aug 2022 – Dec 2022

Commercial Bank of Ceylon, Main Branch Wellawatta

- Handled lower-counter operations, including account opening, debit card issuance, and E-Remittances, ensuring compliance and customer satisfaction

Education

BSc (Hons) in Computer Engineering,

2021 – 2026

The Open University of Sri Lanka

Current GPA: 3.58

Focus: Embedded Systems, AI/ML, IoT, Digital System Design, Computer Vision, Processor Design

Featured Projects

Automatic Number Plate Recognition (ANPR) and

Present

Vehicle Park Access Control System, Tech Stack: Python,

ReactJS, Computer Vision, OCR, YOLO, MQTT

- Developing an end-to-end ANPR system, focusing heavily on OCR optimization and system architecture to ensure accurate, real-time plate reading.
- Actively training and fine-tuning models to recognize highly variable localized data, specifically targeting the full spectrum of Sri Lankan number plates, including older, non-standard, and new custom formats.

SecureGate - IoT Access Control System, Tech Stack:

2026

EasyEDA, FastAPI, ReactJS, PostgreSQL, MQTT, WebSocket,

Nginx, CI/CD

- Designed custom PCB hardware and managed the complete end-to-end deployment lifecycle, including cloud hosting and CI/CD pipelines.
- Architected a multi-VM infrastructure utilizing an Nginx-powered load balancer, a dedicated MQTT broker, and a unified Application VM hosting both the frontend and backend.
- Developed a robust FastAPI backend and React frontend, deployed using Uvicorn and PM2.
- Developed secure end-to-end communication between the FastAPI backend, React frontend, and ESP32 nodes by implementing RSA and dynamic key exchange protocols.

Fuzzy Matrix Processing Unit (University Project) [🔗](#),

2025

Tech Stack: VHDL | Xilinx Vivado | FPGA (Artix-7) | Digital

System Design

- Architected a dedicated hardware accelerator using VHDL to perform Fuzzy Matrix operations.
- Validated design integrity through testbenches and post-synthesis timing simulations in Xilinx Vivado.
- Implemented and verified the synthesized design on a Nexys A7 Development Board (Artix-7 FPGA), demonstrating real-world hardware functionality.

Multi-Unit Environmental & Agricultural Monitoring

2025

System (On Going)  , Tech Stack: ESP32 | Pi Zero |

Embedded C | Python | MQTT | SDI-12

- Architected a wireless sensor network with three coordinated ESP32 nodes for real-time soil (TEROS 10/22), environmental, and weather (ATMOS 41) monitoring.
- Designed a robust data pipeline where edge nodes serve sensor data via ESPNow, consumed by a central aggregator for upload via a MQTT Broker.
- Programmed a self-sufficient weather station capable of parsing complex SDI-12 protocols and logging timestamped data.
- Developed the system's backend architecture using Python FastAPI and created the frontend dashboard using Next.js.

Portable GHG Analyzer for Agricultural Field

2025

Research  , Tech Stack: ESP32 | Embedded C | SPIFFS

- Built a standalone, keypad-operated device for precision Greenhouse Gas measurement, significantly improving the efficiency of agricultural field research.
- Implemented a dual-file system using SPIFFS for persistent calibration data storage and an SD card for logging individual sample data.
- Developed automated control algorithms for air circulation and stabilization, ensuring high-fidelity readings in varying outdoor conditions.

Helmholtz Cage - Nano Sterlite Orbital Magnetic Field

2024

Simulation Device  , Tech Stack: Embedded C | ESP32 |

Python | PID Control | WebSockets

- Implemented precise current control with an embedded web dashboard hosted via SPIFFS directly on the ESP32 for low-latency remote monitoring.
- Developed a self-balancing, frictionless air-bearing platform using a PID algorithm to simulate zero-gravity orbital conditions.
- Integrated and managed industry-level sensors and actuators
- Implemented FreeRTOS for precise task scheduling to manage magnetic field generation and sensor feedback loops

✦ Extra Curricular Activities

Student Volunteer  , USBus Mobile American Space Program 2025

President  , ENOSPIRE OUSL 2025 Entrepreneurship Day Organizing Committee

Organizing Committee Member  , TalentLyft '25 Career Fair, OUSL

Technical Team Lead  , TalentLyft '23 Career Fair, OUSL

Volunteer Teacher & Committee Member, SVP Educational Institute, Kalutara

🔗 References

Prof. Channa J Kumarage, Computer Information Systems Professor,
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